

Program 07: Deploy a web application to kubernetes using Minikube

Structure of the program:

Nginx-deployment.yaml

Steps:

1. Start the minikube : minikube start
2. Check the status of the minikube : minikube status
3. It shows the following status

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
❗ Failing to connect to https://registry.k8s.io/ from both inside the minikube container and host machine
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🔄 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌞 Enabled addons: default-storageclass, storage-provisioner
🏡 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

- 4 . If some the services didnt start then run the following command

minikube delete

minikube start - - driver=docker

5. Build the deployment yaml file

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~$ cd DevOps/lab/p7
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p7$ nano nginx-deployment.yaml
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p7$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p7$
```

6. Apply the nginx-deployment.yaml file

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

7. Check the status

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    1/2     2            1           40s
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p7$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
nginx-deployment-6f9664446b-8wwfp   1/1     Running            0          50s
nginx-deployment-6f9664446b-dxpft   0/1     ImagePullBackOff    0          50s
```

8. Expose the deployment as service

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p7$ kubectl get services
NAME         TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes   ClusterIP     10.96.0.1     <none>         443/TCP          3m48s
kubia        NodePort      10.104.189.20 <none>         8080:32388/TCP   3d23h
kubia-http   LoadBalancer 10.105.72.15  <pending>      8080:31683/TCP   3d23h
nginx-deployment NodePort      10.111.40.57  <none>         80:31639/TCP     10m
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p7$ kubectl delete service nginx-deployment
service "nginx-deployment" deleted from default namespace
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p7$ kubectl get services
NAME         TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes   ClusterIP     10.96.0.1     <none>         443/TCP          4m15s
kubia        NodePort      10.104.189.20 <none>         8080:32388/TCP   3d23h
kubia-http   LoadBalancer 10.105.72.15  <pending>      8080:31683/TCP   3d23h
nginx-deployment NodePort      10.100.195.29 <none>         80:31588/TCP     3s
```

9. Access the web application

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31588

Opening service default/nginx-deployment in default browser...

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p7$ Gtk-Message: 10:14:26.561: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
```

10. View the output in the web browser

